

Multiple Choice (10 marks)

1. Statement 1 | Every ideal in a ring is a subring of the ring. Statement 2 | Every subring of every ring is an ideal of the ring.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. In the xy-plane, the graph of $x^{\log y} = y^{\log x}$ is
 - (a) the open first quadrant
 - (b) a closed curve
 - (c) a ray in the open first quadrant
 - (d) a single point
3. Find the degree for the given field extension $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ over $\mathbb{Q}$.
 - (a) 0
 - (b) 4
 - (c) 2
 - (d) 6
4. Find the generator for the finite field $\mathbb{Z}_7$.
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. Which of the following is true?
 - (a) Every compact space is complete
 - (b) Every complete space is compact
 - (c) Neither (a) nor (b).
 - (d) Both (a) and (b).

True / False (10 marks)

Answer True or False

6. True or False: The characteristic of the ring $2\mathbb{Z}$ is 0, indicating that it contains elements of infinite order.
7. True or False: The linear transformation T maps the point $(2, 1)$ to $(1, 6)$.
8. True or False: The equation of the line tangent to the graph of $y = x + e^x$ at $x = 0$ is $y = 2x + 1$.
9. True or False: A complete graph with 10 vertices has 20 edges, connecting each vertex to only one other vertex.
10. True or False: A function is a permutation if and only if it is onto, regardless of whether it is one-to-one.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Solve

$$\frac{2x + 4}{x^2 + 4x - 5} = \frac{2 - x}{x - 1}$$

for x .

12. Discuss the implications of Statement 1 regarding the relationship between continuity, compactness, and uniform continuity. Additionally, analyze Statement 2 on the differentiability of composite functions.

End of Paper